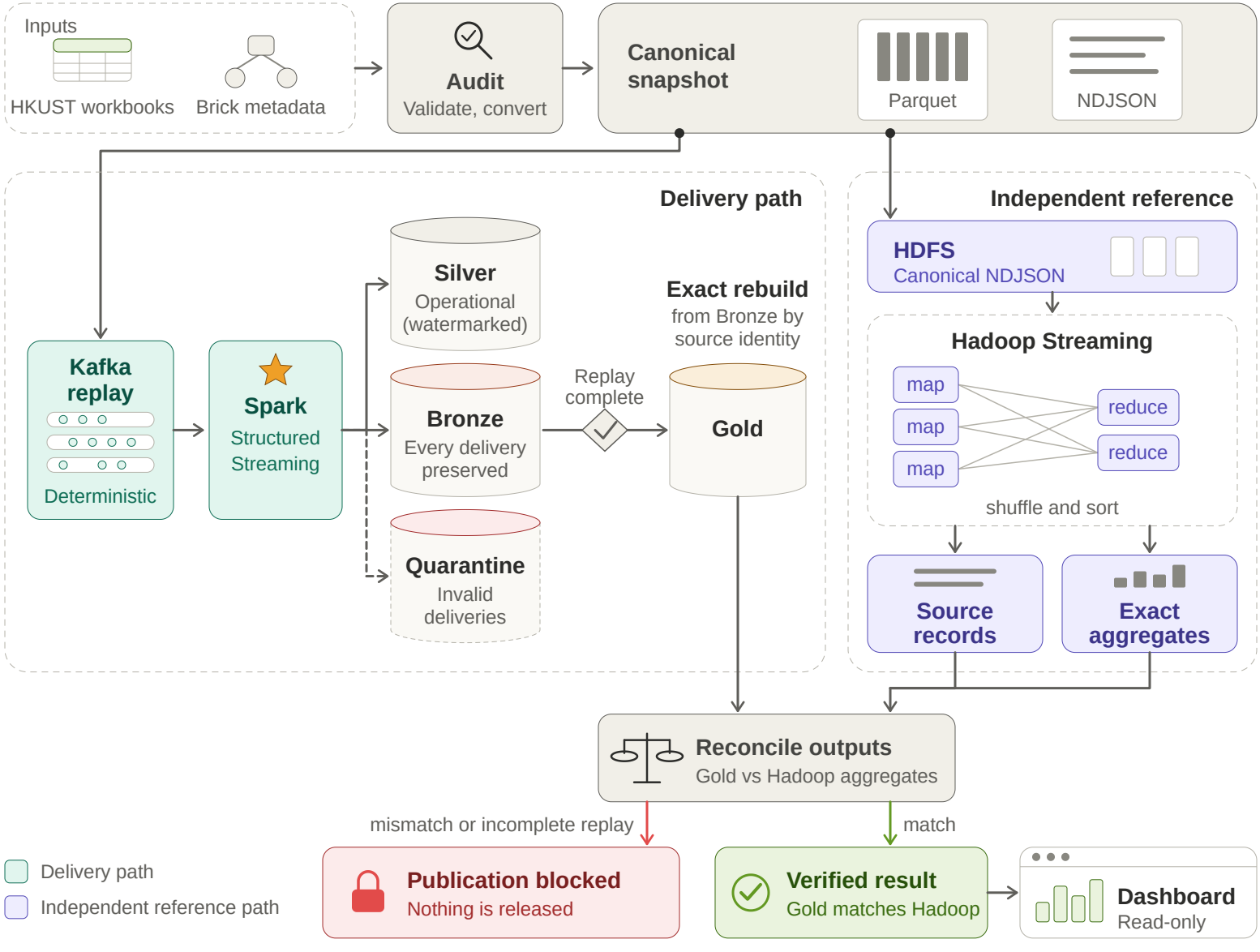

Inputs
HKUST workbooks
Brick metadata
Audit
Validate, convert
Canonical snapshot
Parquet
NDJSON
Delivery path
Independent reference
Kafka replay
Deterministic
Spark
Structured Streaming
Silver
Operational (watermarked)
Bronze
Every delivery preserved
Quarantine
Invalid deliveries
Replay complete
Exact rebuild
from Bronze by source identity
Gold
HDFS
Canonical NDJSON
Hadoop Streaming
map
map
map
reduce
reduce
shuffle and sort
Source records
Exact aggregates
Reconcile outputs
Gold vs Hadoop aggregates
mismatch or incomplete replay
match
Publication blocked
Nothing is released
Verified result
Gold matches Hadoop
Dashboard
Read-only
Delivery path
Independent reference path